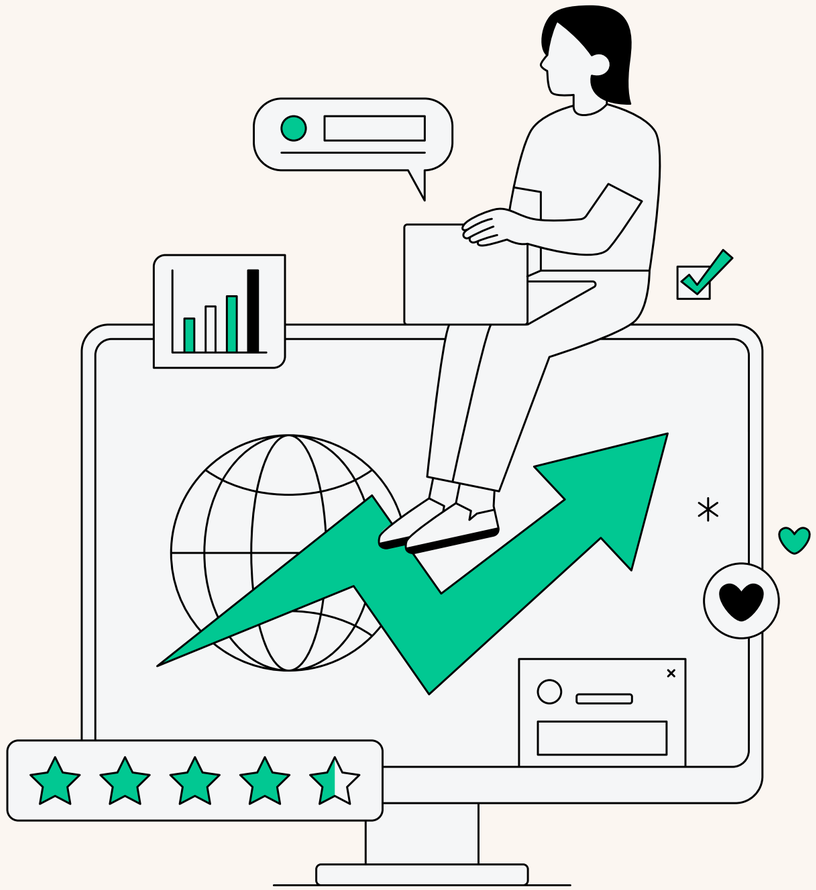
CSEN140

Analyzing Salary Fairness

By: Robert Rossi, Emily Chen, Ryan Vattuone



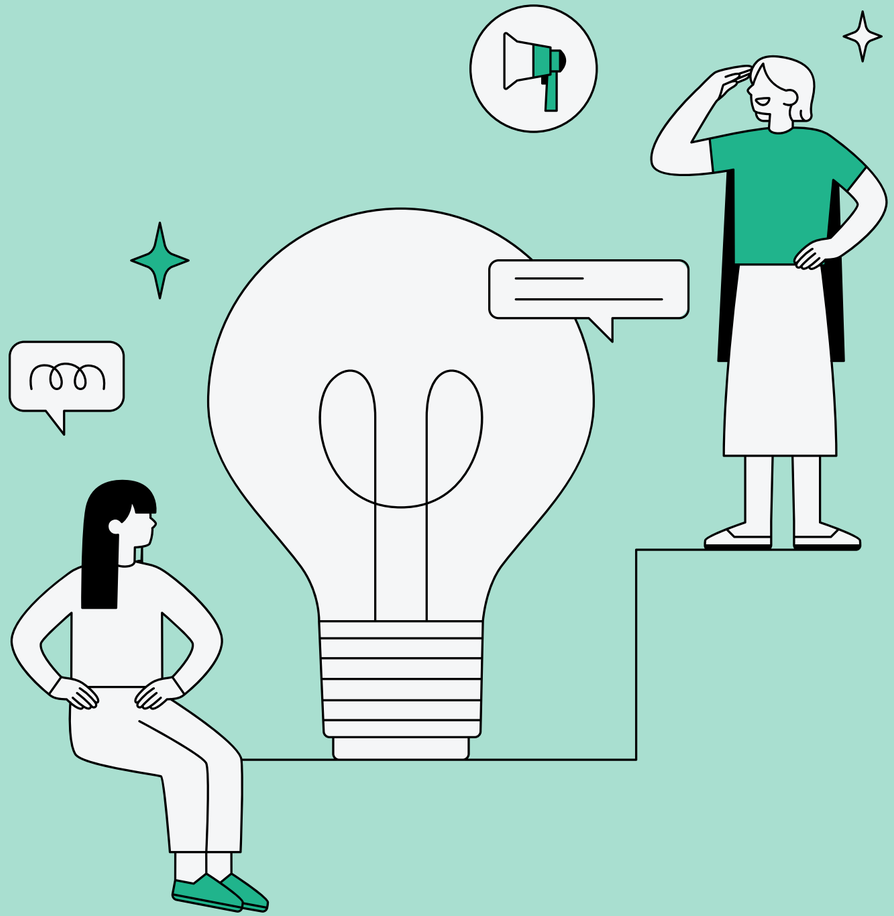
Problem Statement

- Salaries are very different across different companies, locations, and occupations but a **higher salary does not always mean a better purchasing power**.

- Different states are associated with **different costs of living, expenses, and taxes**, meaning that the **same dollar could stretch further** in some states as opposed to others.

- There is a **lack of transparency** on how companies adjust their compensation values from state to state, making it **challenging for job seekers to reliably compare** the **real value** compensation packages across states.

- Our project will analyze salary data at different companies and **determine whether employees are being generously, fairly, or under-compensated** given their salary, occupation, and state.



Datasets

01.

Companies and Jobs Dataset

- Includes: company name, job title, salary information, location, and company attributes
- Size: 100,000+ samples and 10-20 features before processing

02.

BLS Employment Dataset

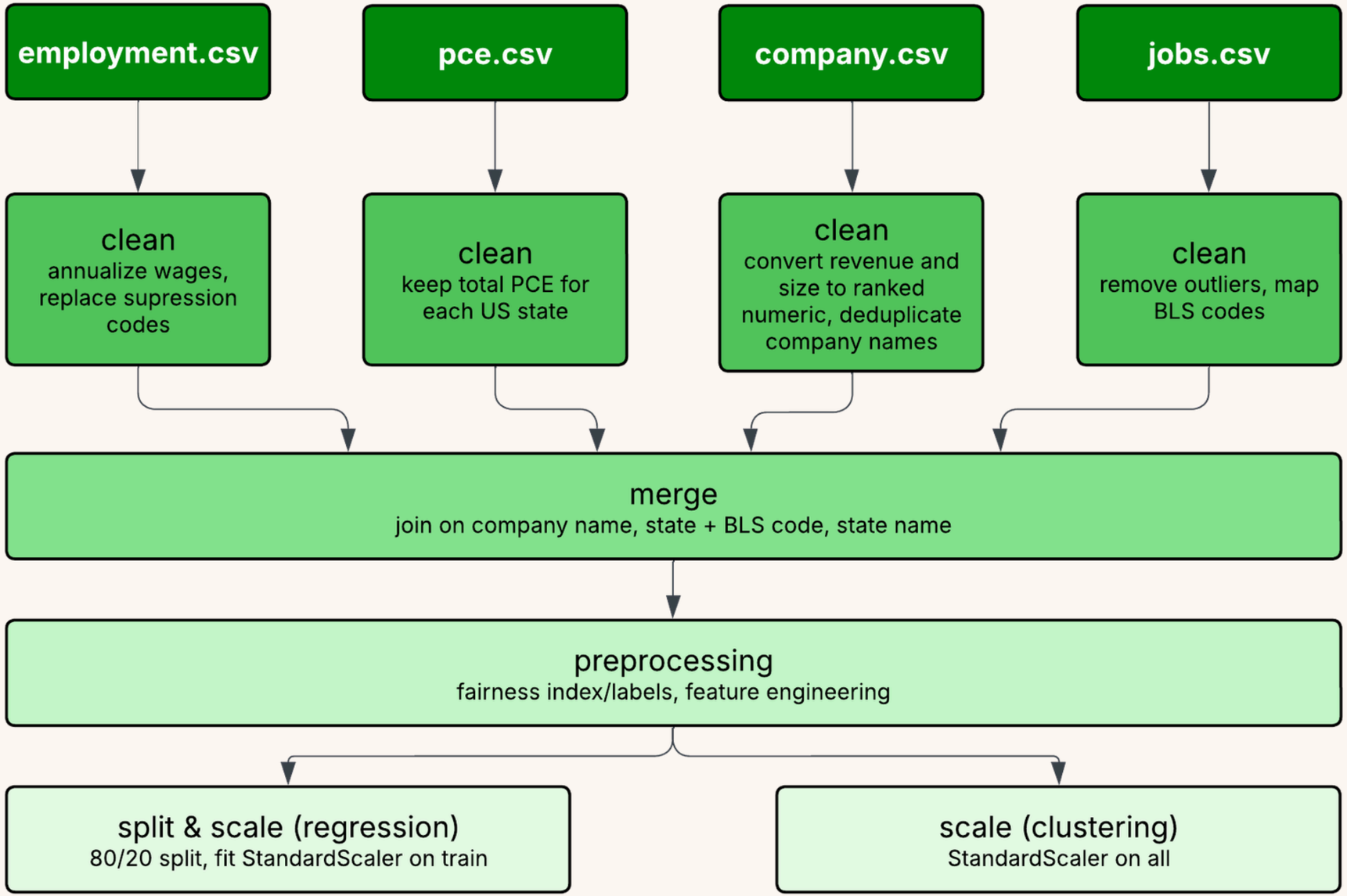
- Includes: area/location, occupation, total estimated number employed (excludes self-employed), mean hourly wage, mean annual wage
- Size: 37,609 samples and 32 features before processing

03.

Personal Consumption Expenditure Dataset

- Includes: state/location, region, description, unit, personal consumption expenditure
- Size: 14,440 samples and 36 features before processing

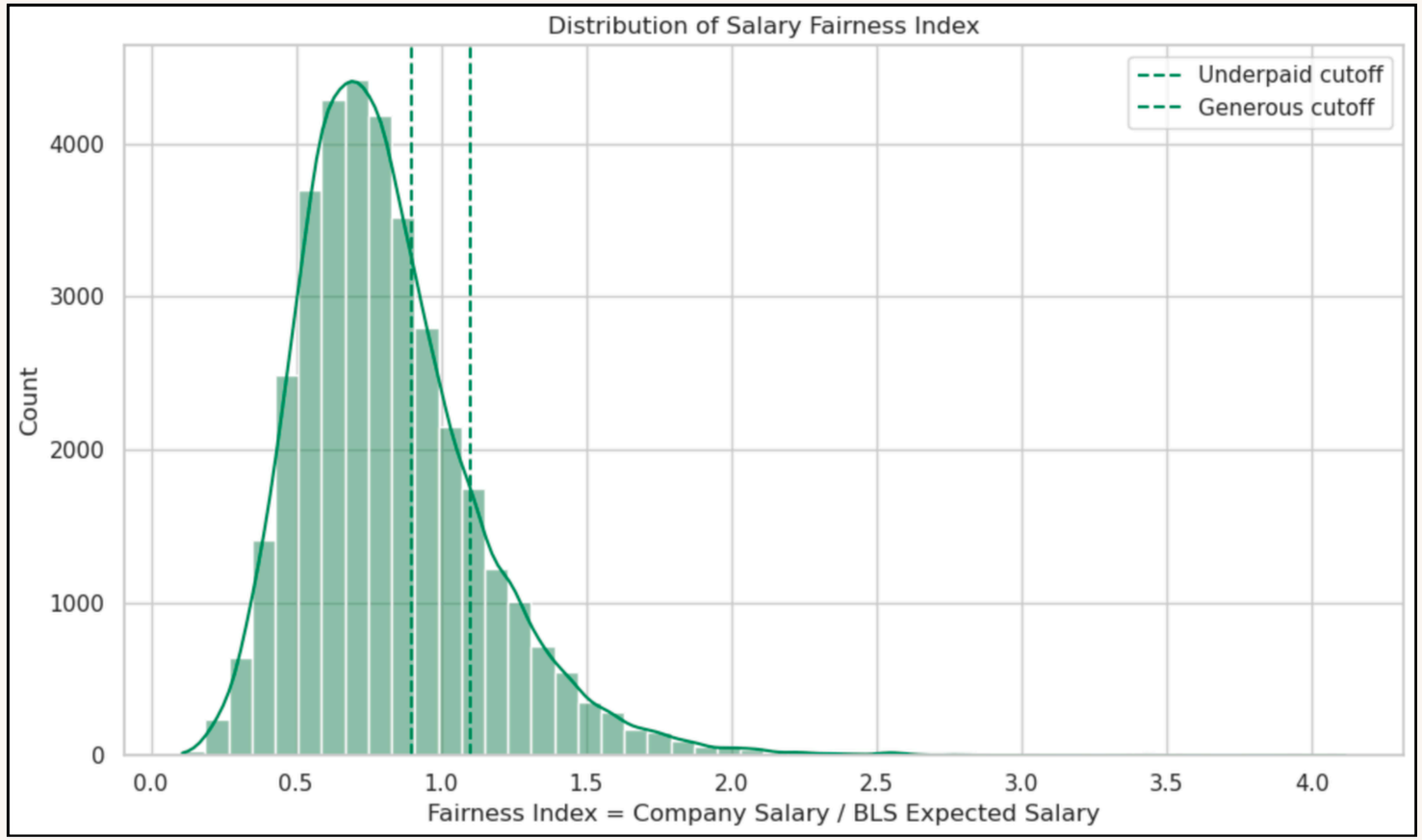
Cleaning & Preprocessing



Fairness Index

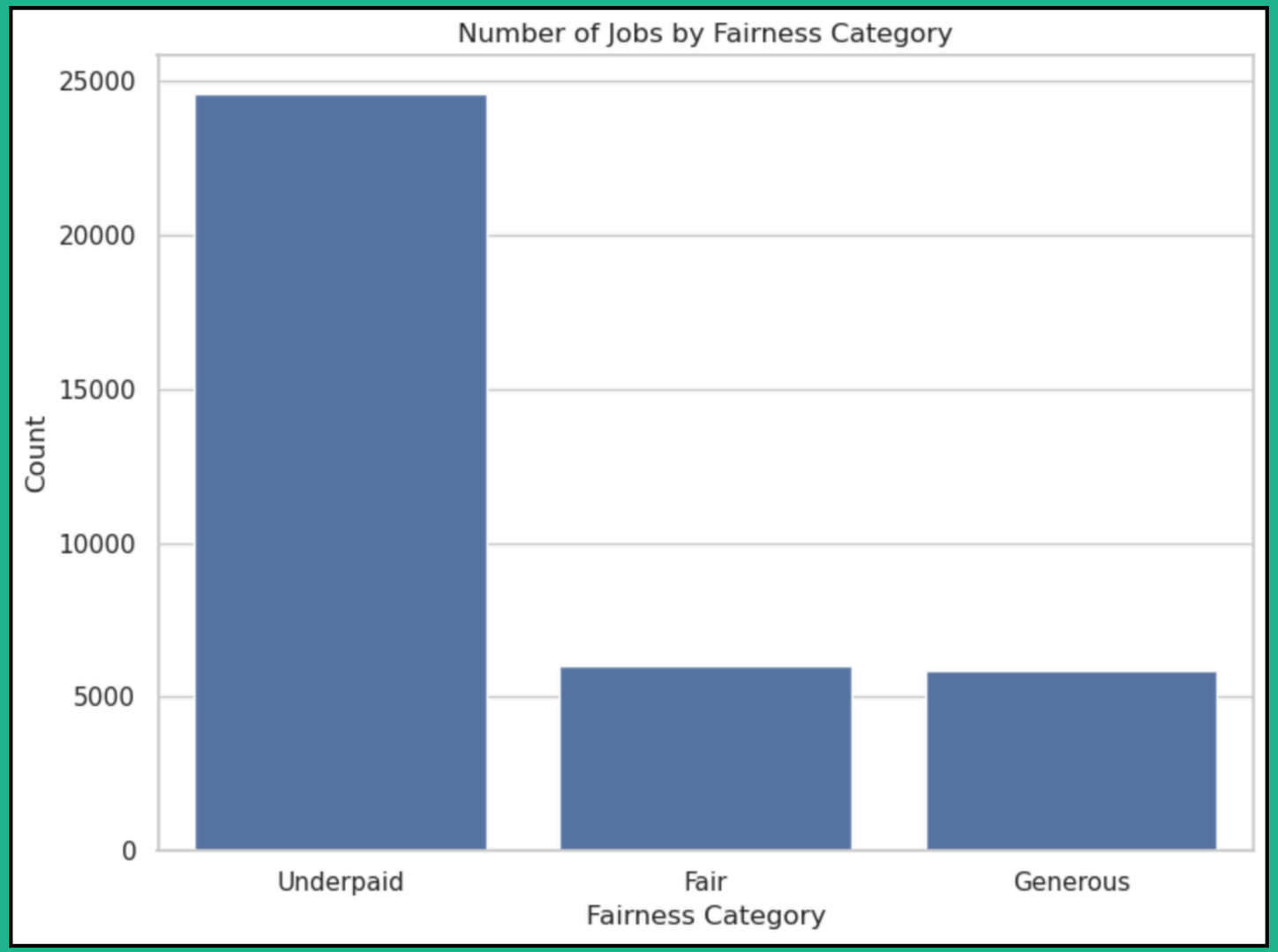
$$\text{Fairness Index} = \frac{\text{Annual Mean}}{\text{BLS Annual Mean}}$$

Fairness Index	Classification
< 0.9	Underpaid
0.9 - 1.1	Fair
> 1.1	Generous



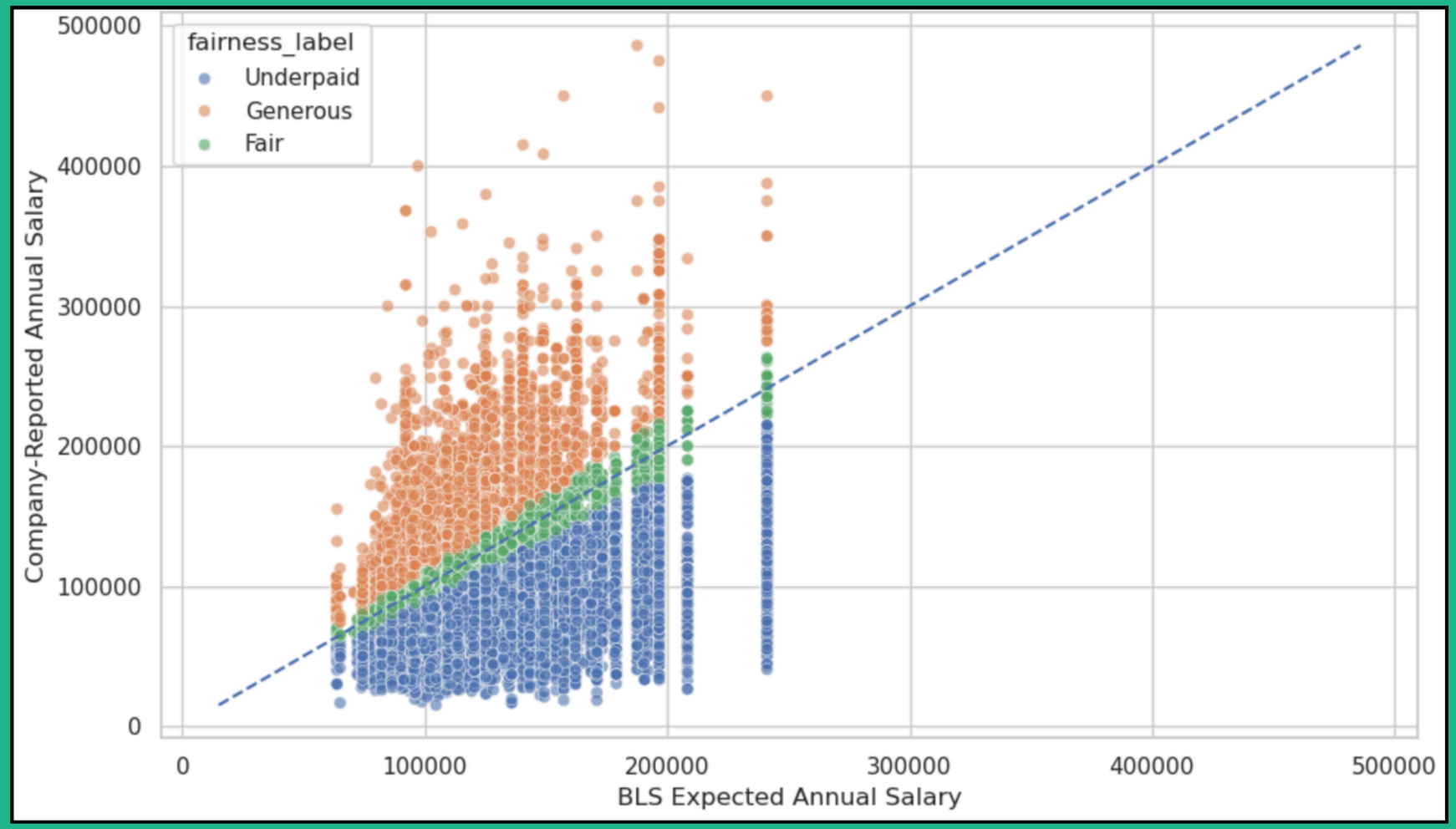
This index compares the listed average salary for a job against the BLS average salary for the same occupation and state.

Salary Fairness Distribution



- Underpaid jobs made up the majority of the dataset: 24,594 jobs, or about 67.5%
- Fair and generous jobs were much smaller groups, each around 16% of the data
- This shows the dataset is skewed toward salaries below BLS expectations

Company Salary vs. Expected Salary



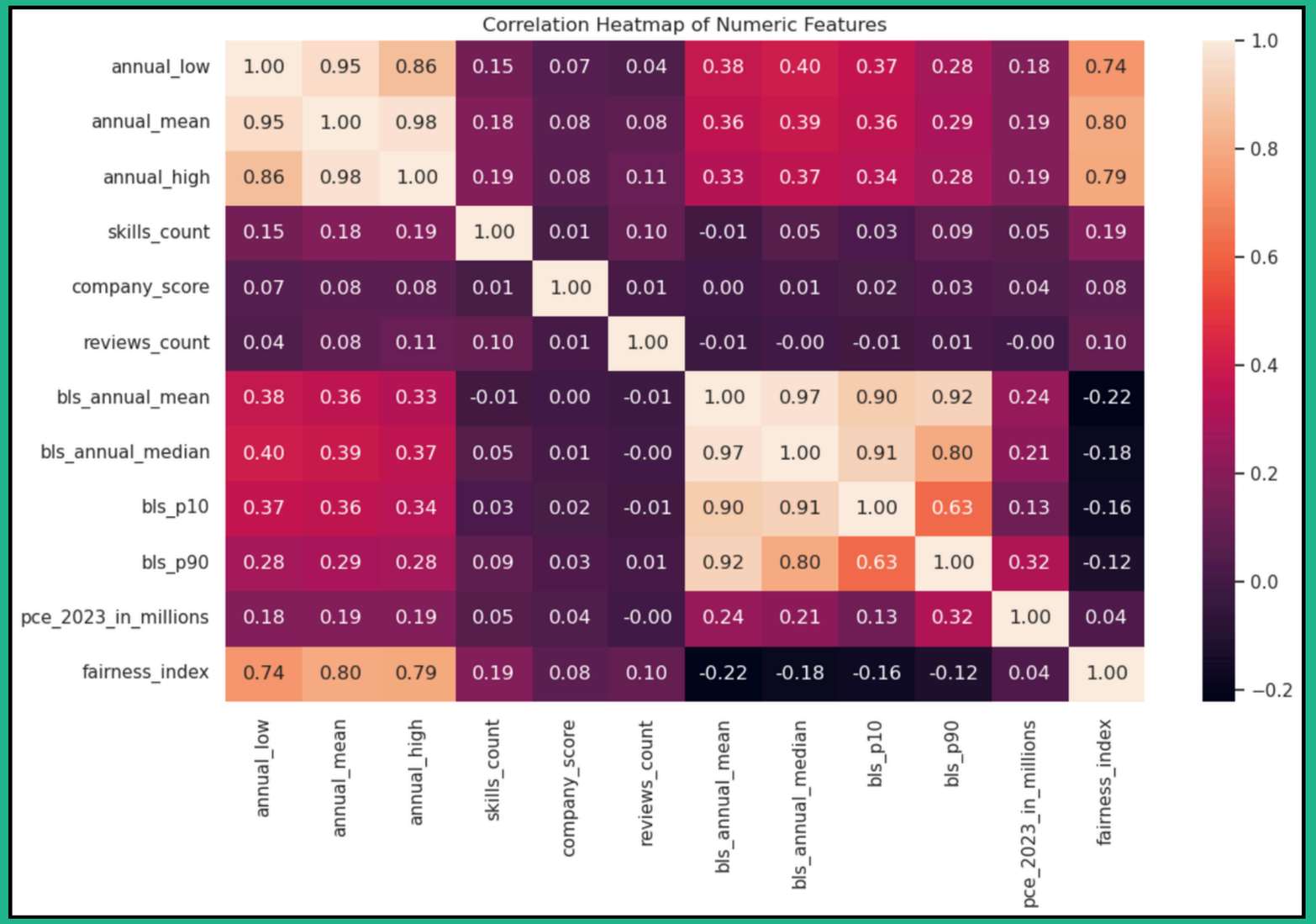
Fair Compensation Line
(Company Salary = Expected Salary)

- Points below the diagonal line represent jobs paid below the BLS expected salary
- Many jobs cluster below the fair compensation line, supporting the underpaid trend from the previous slide
- The large spread shows that jobs with similar BLS benchmarks can still have very different company-reported salaries

Real Examples of Salary Fairness

Job	Company	State	Salary	BLS Salary	Fairness	Classification
Business Analyst	CyberCoders	CT	$95,000	$115,530	0.822	**Underpaid**
Sr. Analyst-Business Intelligence	Verizon	NJ	$99,500	$108,990	0.913	**Fair**
Lead Business Analyst	Stryker	MI	$176,800	$95,710	1.847	**Generous**

Correlation Analysis



- Annual salary measures had the strongest relationship with the fairness index
- BLS wage variables were highly correlated with each other, supporting their use as benchmark salary measures
- PCE showed a weak relationship with fairness, suggesting state-level spending alone does not explain compensation fairness

These EDA results suggest that salary-related features carry the strongest predictive signal, while economic context alone is not enough to explain salary fairness.

Regression Models

Model	RMSE	R^2
Linear Regression	0.245537	0.389738
Ridge Regression	0.245536	0.389743
Lasso Regression	~0.2456	~0.389
Elastic Net	**0.245530**	**0.389771**

- **Elastic Net achieved best overall performance**
- **Regularization provided only minor improvements over linear regression**
- **Similar performance suggests limited overfitting**

Model Comparison

Approach	R^2
Linear Regression	0.389738
Ridge Regression	0.389743
Elastic Net	**0.389771**
PCA + Elastic Net	0.3860
SVD + Elastic Net	0.3835

- **Elastic Net achieved the best overall performance**
- **PCA and SVD reduced dimensionality but slightly reduced accuracy**
- **Original feature set retained the most predictive information**

Clustering

Method	Silhouette Score
K-Means	0.1442
K-Means + PCA	0.1582
DBSCAN	**0.4250**

- **PCA improved K-Means clustering performance by reducing noise and dimensionality**
- **The best K-Means result occurred at k = 2, suggesting the data naturally forms a small number of broad groups**
- **DBSCAN achieved the highest silhouette score and best overall clustering quality**
- **Results indicate that salary fairness data contains non-spherical cluster structures that are better captured by density-based clustering algorithms**

Conclusion

- Developed a salary fairness index using job salary, occupation, and location data
- Successfully integrated and analyzed three large public datasets
- Elastic Net achieved the best regression performance ($R^2 \approx 0.390$, RMSE ≈ 0.246)
- PCA improved clustering performance, while DBSCAN produced the strongest clustering results
- Results demonstrate that salary fairness can be partially predicted using publicly available job and economic data

Overall, public salary, occupation, and cost-of-living data can be combined to provide meaningful insights into compensation fairness across companies and states.



Thank you!

